Global Cultural Narratives around DH Concepts for the Humanities Classroom

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From its earliest years, Digital Humanities has been shaped by debates about whether to prioritize the teaching of technical skills or methodological frameworks—and how best to combine the two. Mahony and Pierazzo (2012) advocated for an integrated approach that balances hands-on technical instruction with theoretical explorations of methods. However, this tools-versus-methods binary has persisted in DH pedagogy discussions. A recent literature review by Georgopoulou et al. (2024) that covers DH pedagogy from 2012 to 2022 frames this debate as one between subjects of study where methods are developed, and fields of application, in which tools are applied. However, this framing risks overlooking a critical question that Alan Liu posed in 2012: "Where is cultural criticism in the Digital Humanities?" Tara McPherson (2012) raised a related concern in her paper, "Why Are the Digital Humanities So White?," with "white" referring to epistemological assumptions (not race). These questions remain pressing, not only for research but also for pedagogy. As the world becomes increasingly globalized, integrating multicultural critique into DH pedagogy is essential. Beyond Western-centric narratives, culturally inclusive methods can address epistemic disjunctions that emerge (Bhattacharyya 2022) as DH expands into new areas. This paper proposes a way to teach DH as an emphatically humanities discipline by moving beyond the tools-versus-methods binary.

I argue for the explicit incorporation of three core moves in humanistic inquiry— historicizing, contextualizing, and analogizing —that are all inherently comparative, into DH teaching. Historicization compares texts and/or ideas across time, contextualization compares them across space, and analogizing connects concepts across both time and space. In the DH classroom, these comparative moves can link DH notions to narratives that resonate with students' cultural and individual interests, curiosities and heritages, weaving these into broader, global narratives. In earlier work, I explored how metaphors from Singapore's political discourse helped students grasp DH concepts. A key dialectic in many DH methods, such as search-space exploration and machine learning, involves balancing freedom and constraint. Generative adversarial networks (GANs), for instance, balance expansion and limitation through generative proliferation and adversarial restriction. I used Anthony Giddens's notion of agency versus structure to relate this dialectical opposition to Singapore's balancing act between Western-style liberal democracy and neo-Confucian notions of stability (Bhattacharyya 2023).

In the present (paper, I draw on my experience teaching Digital Humanities in the Humanities Program at Yale University, especially in a senior-level course I designed, "AI as a Global Cultural Artifact." This course, explores how concepts central to DH, such as approximating semantics in text corpora, can be related to global cultural narratives through historicization, contextualization, and analogizing. With over 44% of Yale's Class of 2025 speaking a language other than English at home (YaleNews 2021), the course provided a good environment for testing these pedagogical strategies. Building on Kitchens (2009), I propose that the educational strategies discussed align with John Dewey's concept of "interplay" between "objective" or "environing" conditions (Kitchens's term) and "internal" conditions (John Dewey's term). This dynamic interaction, as Dewey (1997) describes, forms what he calls a situation," offering a compelling framework for incorporating these strategies into a situationist paradigm.

Fig. 1. Analogizing vector semantics in textual DH by reference to ancient Indian creation myth

Given space constraints, I limit my discussion to vector semantics as pedagogical example. Vector semantics holds significant conceptual value for Digital Humanities pedagogy today, providing a theoretical basis for understanding how language models represent and approximate meaning in the world (Liu 2024). As a native of India—a cultural palimpsest shaped by myth from the clas-

sical Sanskrit tradition and, later, by the culture of the broader Islamicate world—I found it useful to connect vector semantics to these two cultural contexts. Two illustrations follow, taken from my slide deck for the aforementioned course. They illustrate the application of vector semantics to Vedic myth (Fig. 1) and to the Islamicate intellectual tradition (Fig. 2, 3 and 4).

In Fig. 1, syntagmatic and paradigmatic relations are mapped using a matrix—a fundamental semiotic tool—to identify ontological categories (matrix columns) as paradigms. These paradigms correspond to words clustered in vector hyperspace, where co-occurrence relationships define semantic axes (see Gavin 2018 and Heuser 2017 for detailed explanations). The matrix rows, representing word relationships, align with syntagms. Drawing on Roberto Calasso's interpretation of Vedic myth (1998), students are able to appreciate how myths symbolically emphasize the ontological primacy of semantics over individual subjectivity. In Fig. 2, 3 and 4, I historicize using Tenen's analysis (2024) of the medieval Arabic zairajah , a device used to generate ideas combinatorially from words by mechanical means, and Ramon Llull's Christian mystical adaptation, influenced by Jewish Kabbalistic thought, of the idea underlying the zairajah . These contraptions attempted to generate only meaningful and "valid" phrase combinations, paralleling modern semantic filtering in syntagmatic relationships.

Fig. 2. Historicizing vector semantics by means of its connections with the Arab-Islamicate world

Fig. 3. Fleshing out and extending the historicization initiated in Fig. 2

To contextualize this learning experience, students were provided with museum-based sources,such as online encyclopedias and artifact databases, as well as access to supplementary readings such as Forcada (2008) and Gamarra (2015). Through perusal of these materials and by means of class discussion, students come to discover how discursive influences such as the use of concentric circles (Fig. 4) that may have been at play, not only for divination but also for city planning, shaped the combinatorial imagination within this epistemic framework (Forcada 2008; Gamarra 2015). By connecting vector semantics to these global cultural narratives, students can move beyond seeing DH as purely technical or even purely methodological. Instead, they can engage with DH as a field that links together global cultural histories, epistemological frameworks, and symbolic traditions. I contend that this kind of approach can transcend the restrictive binary of tools-versus-methods dichotomy, and can help foster a DH education that is an emphatically humanistic education in a fuller sense of the term.

Fig. 4. Contextualizing the learning by discursive connections:
Concentric circles with symbolic markings show up in multiple contexts within this episteme.

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